

FoodLoop

Team Number	CS11-184
Team Name	HyperNova
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Project Subtrack	Platforms

1. PROBLEM DEFINITION & BACKGROUND

1.1. Problem Statement

Despite millions of people facing food insecurity daily, vast quantities of edible surplus food from restaurants, supermarkets, and households are discarded due to the absence of an efficient, trustworthy, and economically viable redistribution system. This disconnect not only deepens hunger and social inequality but also accelerates environmental damage through methane-producing landfill waste, while small food businesses and vulnerable communities continue to lack a unified platform that makes nutritious food both accessible and affordable.

1.2. Problem Background

Food waste and food insecurity present a critical global and local contradiction. While the FAO reports say that 1.3 billion tons of food wasted annually, over 733 million people face chronic hunger. In Sri Lanka, this is intensified by a 40% post-harvest loss rate and rising urban food insecurity.

Core Challenges:

- **Business Inefficiency:** Supermarkets and restaurants lack real-time mechanisms to discount or redirect surplus, leading to mass disposal of near-expiry stock.
- **Market Fragmentation:** Small-scale entrepreneurs remain excluded from the formal economy due to a lack of accessible, trusted marketplaces.
- **Corporate Accountability:** Large firms struggle to fulfill CSR goals without verifiable data or transparent donation tracking.
- **Environmental Crisis:** Rotting food in landfills produces methane, a greenhouse gas 80 times more potent than CO_2 . Conversely, every ton of food diverted prevents approximately 1.9 tons of CO_2 equivalent emissions.

Current solutions are fragmented, manual, and unable to scale. FoodLoop bridges this gap by offering an AI-powered circular ecosystem that unifies surplus management, verified social aid, and environmental accountability into a single, intelligent platform.

1.3. Research Background

The technical and social architecture of FoodLoop is validated by global research and local data, addressing a systemic failure in current food management.

1. Global & Local Food Crisis

- **Scale of Waste:** The UNEP Food Waste Index (2024) reports 1.05 billion tonnes of food wasted annually. Digital intervention at the point of surplus is cited as a primary mitigation strategy.
- **The Hunger Paradox:** While millions face hunger (FAO, 2023), Sri Lanka faces a 30 - 40% post-harvest loss (HARTI, 2020). Recent economic shifts have specifically heightened food insecurity among urban residents and students (IPS, 2022).

2. Technology-Driven Solutions

- **AI & Dynamic Pricing:** Research in the Journal of Cleaner Production (2020) and McKinsey (2022) confirm that AI-driven markdown strategies reduce retail waste by up to 35% while maximizing ROI.

- Blockchain Transparency: Studies by Kshetri (2018) and IBM Food Trust demonstrate that blockchain creates the immutable records necessary to build consumer trust and donation accountability.
 - Smart Logistics: The World Resources Institute (2021) identifies AI-powered routing as a critical tool, reducing redistribution emissions by 20% and improving delivery success by 30%.
3. The Enterprise ESG Gap
- Verification Crisis: A Deloitte (2023) survey found that 75% of leaders prioritize ESG data, yet fewer than 30% have access to real-time, verified metrics. This confirms the market demand for FoodLoop's Blockchain-backed CSR Suite.

Research Focus	Key Metric / Finding	FoodLoop Feature
UNEP / FAO	1.05B tons wasted; 783M hungry	Unified Marketplace
McKinsey	Real-time AI triggers = Highest ROI	Dynamic Discounting
WRI	Smart routing = 20% less emissions	AI Smart Routing
Deloitte	70% lack verified impact data	CSR/ESG Dashboard

1.4. In-Scope & Out-Scope

FoodLoop establishes clear boundaries to maintain its focus on an AI-driven circular ecosystem while defining its operational limits.

Core Functionalities (In-Scope)

- Smart Listing & Safety:** Employs AI "Snap & List" for rapid donations and computer vision to screen for food spoilage before listings are published.
- Dynamic Commerce:** Features a C2C/B2C marketplace for small vendors and an AI-driven dynamic discounting engine for supermarkets to manage near-expiry goods.
- Automated Redistribution:** Automatically converts unsold discounted items into donations for verified NGOs or low-income users.
- Verified Social Aid:** Includes an eligibility module to categorize students and low-income users for exclusive subsidies and priority claims.
- Blockchain Transparency:** Uses an immutable ledger to track food from donor to recipient, generating digital "Impact Receipts" and CSR reports.
- Intelligent Logistics:** Utilizes real-time maps and AI smart routing to match food volumes with the optimal vehicle types.
- Predictive Analytics:** Incorporates "Demand Sensing" that uses weather and local event data to forecast waste and trigger proactive discounts.

- **Engagement & Support:** Provides a 24/7 AI chatbot for user assistance and a gamified reward system to encourage donor participation.

Boundaries (Out-of-Scope)

- **Physical Infrastructure:** The platform does not own warehouses, cold storage, or distribution centers.
- **Direct Fulfillment:** FoodLoop coordinates volunteers but does not employ its own delivery fleet or manufacture food.
- **Inventory Management:** It does not replace supermarket ERP systems and is restricted to managing near-expiry listings only.
- **Financial & Advisory Services:** No credit/lending facilities or personalized medical/nutritional advice are provided.
- **Operational Limits:** Initially restricted to the Sri Lankan market and requires an active internet connection (no offline/SMS mode).
- **Legal Enforcement:** While automated checks assist with safety, final legal compliance remains the responsibility of individual sellers and donor

1.5. User Pain Points

FoodLoop resolves critical systemic failures that manual or fragmented methods cannot address.

- **Food Donors (Restaurants & Individuals):** High operational friction and slow manual entry lead to surplus disposal. A lack of verifiable proof and rewards creates liability fears and donor fatigue.
- **Supermarkets & Retailers:** Reactive, labor-intensive expiry management and a lack of demand forecasting lead to financial losses and avoidable overstocking. Currently, no automated "safety net" exists to redirect unsold items to donation channels.
- **Small Food Businesses & Home Sellers:** These entrepreneurs face informal barriers like inconsistent visibility and a lack of structured payments on social media. Without platform-level safety verification, they suffer from a credibility crisis with potential buyers.
- **NGOs & Volunteer Organizations:** Distribution planning is hindered by unpredictable supply and the absence of real-time visibility. Inefficient logistics and manual reporting burdens increase fuel costs and slow down impact verification for CSR.
- **Vulnerable Groups (Students & Low-Income):** These users often face social stigma and miss time-sensitive opportunities due to the lack of a dignified, technology-enabled pathway and proximity-based alerts.
- **Corporate Entities:** Firms struggle with a significant "Data Gap," finding it slow and costly to aggregate fragmented data into verified, real-time ESG metrics for regulatory reporting.

2. Proposed Solution

2.1. One-Line Pitch

FoodLoop is an AI-powered circular food economy platform that unifies surplus donors, small businesses, and vulnerable communities through intelligent automation and blockchain-verified transparency to eliminate food waste.

2.2. Solution Overview

FoodLoop integrates commerce, social aid, and environmental accountability into a single ecosystem:

- **Commerce Layer:** Includes a dual C2C/B2C marketplace and an AI engine for dynamic discounting on near-expiry goods that automatically converts unsold stock into donations.
- **Social Impact Layer:** Provides verified subsidies for low-income individuals and students, alongside a Live Impact Map with AI-optimized routing for NGO redistribution.
- **Trust & Accountability Layer:** Utilizes blockchain for immutable food traceability and provides corporate partners with verified ESG reports and impact widgets.
- **Integrated AI & Gamification:** Features "Snap & List" food identification, 24/7 chatbot support, and a milestone-based reward system for consistent donors.

2.3. Purpose & Impact

Designed as a self-sustaining business platform, FoodLoop replaces manual redistribution with an intelligent circular economy.

- **Stakeholder Impact:** Donors benefit from reduced friction via AI listing ; retailers slash financial losses through automated markdowns ; small businesses gain a credible, safety-verified marketplace ; and vulnerable communities receive dignified, prioritized access to affordable food.
- **Environmental Impact:** By diverting organic waste from landfills, the platform prevents methane emissions and supports national climate targets.

2.4. Ethical & Social Impact

The platform follows a "**Privacy-by-Design**" and "**Ethics-First**" architecture:

- **Privacy & Equity:** Sensitive user data is encrypted and anonymized while AI tools ensure digital inclusion for all literacy levels.
- **Safety & Transparency:** AI computer vision screens for spoilage to protect health , while blockchain creates immutable records for radical transparency.
- **Community & Climate:** Discrete subsidies remove "charity stigma" , and optimized logistics align with UN Sustainable Development Goals.

2.5. Uniqueness & Special Features

FoodLoop defines a new category by unifying disparate waste-management elements into one loop:

- **Zero-Friction AI:** Uses Computer Vision to identify food and estimate quantity from a single photo.
- **Zero-Waste Pipeline:** Automatically bridges the gap between retail commerce and charity through dynamic conversion of unsold stock.
- **Proactive Demand Sensing:** Monitors weather and local events to forecast surplus risk and suggest preemptive discounts.
- **Optimized Logistics:** An AI dispatch system matches vehicle types to food volume to minimize carbon footprints.
- **Structured Dignity:** The only platform with a verified eligibility layer embedded directly within the main marketplace architecture

3. Market & Industry Analysis

3.1. Target Audience & Market Size

FoodLoop serves a multi-sided ecosystem where every participant acts as a value-driver for the others. The platform taps into a massive, underserved market spanning commercial, social, and corporate sectors.

1. Primary Market Segments

- Food Donors (Restaurants & Catering): * Scale: ~15,000 registered establishments; 150k+ annual events.
 - Need: Rapid, "zero-friction" listing via AI Snap & List to fit intense operational workflows.
- Supermarkets & Retailers:
 - Scale: 2,000+ modern outlets (e.g., Keells, Cargills).
 - Need: Automated inventory recovery. The addressable loss from near-expiry waste is estimated in the hundreds of millions of rupees annually.
- Small Businesses & Home Sellers:
 - Scale: Tens of thousands in the growing informal economy.
 - Need: A formal marketplace providing trust, secure payments, and safety verification to scale beyond social media.
- NGOs & Volunteers:
 - Scale: 1,500+ registered NGOs and thousands of religious institutions.
 - Need: Real-time visibility and AI Smart Routing to serve more people with fewer resources.
- Vulnerable Communities (Students/Low-Income):
 - Scale: 120,000+ university students and ~3 million individuals below the poverty line.
 - Need: Dignified, priority access to subsidized and donated food via mobile alerts.

2. Secondary & High-Value Segments

- Corporate Entities (ESG Clients):
 - Scale: 290+ CSE-listed companies and thousands of private firms.
 - Need: Blockchain-backed ESG data for formal sustainability reporting. This is the platform's highest revenue-potential segment.
- Individual Households:
 - Need: A simple, altruistic outlet for surplus food from festivals and family gatherings.

3.2. Market Potential & Importance

FoodLoop enters the market at a rare convergence of escalating food insecurity, rapid digital commerce adoption, and mandatory corporate ESG obligations.

1. Global & Local Industry Trends

- **Surging Food Tech Market:** The global food waste management market is projected to reach USD 35.5 billion, with technology-driven redistribution being the fastest-growing sub-segment.
- **AI as the Standard:** With AI in the food sector growing at a 45.7% CAGR, FoodLoop's integration of computer vision and demand sensing ensures it remains at the industry's cutting edge.
- **Mandatory ESG Disclosures:** Regulatory shifts (like ISSB standards) are moving ESG reporting from voluntary to mandatory. This creates an urgent corporate need for blockchain-backed, verifiable data.
- **Digitization of Informal Economy:** Post-2022, Sri Lanka has seen a massive shift of home-based food businesses to digital platforms, providing a ready-made, tech-savvy user base for the marketplace.

2. The Sri Lankan Opportunity

- **Economic Urgency:** The residual impact of the 2022 crisis has heightened demand for affordable food and efficient waste management.
- **High Digital Penetration:** Over 50% mobile internet penetration and dense urban centers (Colombo, Kandy) provide the ideal infrastructure for rapid platform scaling.
- **First-Mover Advantage:** Currently, no competitor in Sri Lanka unifies donation, commerce, retail AI, and ESG services. FoodLoop is positioned to define the category.

3. Long-Term Sustainability

- **Multi-Sided Network Effects:** As more donors join, the supply increases, attracting more NGOs and recipients, which in turn makes the blockchain data more valuable for corporate subscribers.
- **Diversified Revenue Streams:**
 - B2C/C2C: Marketplace transaction commissions.
 - B2B: Supermarket dynamic discounting service fees.
 - Enterprise: High-value, recurring Premium CSR subscriptions.
- **Strategic Data Asset:** As the platform scales, the anonymized data on food surplus patterns and redistribution efficiency becomes a valuable asset for government agencies and researchers.
- **Scalability:** The architecture is built for geographic expansion into neighboring markets like India and Bangladesh, where similar structural food challenges exist at a larger scale.

4. Business & Financial Model

4.1. Business Revenue Model & Cost Structure

The Business Model Canvas - FoodLoop				
Key Partnerships • NGOs & Volunteer Orgs (1,500+ in Sri Lanka) • Supermarket Chains (Keells, Cargills) • Universities & Student Ambassador Networks • AWS Activate & Google for Startups • UNDP / FAO / WFP Accelerators • Lanka Angel Network • Polygon Blockchain	Key Activities • AI Snap & List food listing • AI Dynamic Discounting • Blockchain ledger recording • Smart route optimisation • Demand sensing & forecasting • Gamification & rewards system	Value Propositions • Zero-friction AI Snap & List (listing in under 60 secs) • Automated zero-waste retail pipeline • Blockchain-verified trust & digital Impact Receipts • Proactive Demand Sensing (weather + local events) • Smart vehicle-matched AI logistics routing • Dignified subsidised access for vulnerable communities • Real-time ESG dashboard for corporate partners	Customer Relationships • 24/7 AI chatbot support • Gamified donor milestones & digital badges • Blockchain Impact Receipts • Founding Partner badges • Student Ambassador programme	Customer Segments Primary: • Food Donors (15,000+ restaurants) • Supermarkets & Retailers (2,000+ outlets) • Small Businesses & Home Sellers • NGOs & Volunteers (1,500+ organisations) • Vulnerable Communities (students, low-income) Secondary: • Corporate ESG Clients (290+ CSE companies) • Individual Households
Key Resources • AI/ML models (CV, demand sensing) • Polygon blockchain infra • React.js / Node.js / FastAPI / MongoDB stack • Verified user database • Anonymised impact data			Channels • Mobile app (iOS & Android) • Web platform (React.js) • Social media & SEO • University ambassador networks • NGO & retail partnerships	
Cost Structure • Cloud infrastructure: AWS / Render / Vercel hosting fees • AI/ML model training & maintenance (computer vision, demand sensing) • Blockchain transaction & Polygon network fees • Mobile & web app development and ongoing maintenance • Marketing, community outreach & Student Ambassador programme • Payment gateway integration fees Legal & compliance costs			Revenue Streams • Marketplace Commissions: % fee on every C2C/B2C transaction by home sellers • Retail Integration Fees: recurring B2B subscription for AI Dynamic Discounting & Demand Sensing • Premium CSR Subscriptions: tiered annual packages for corporates needing blockchain-verified ESG reports • Sponsored Impact Campaigns: corporate-funded food drives (e.g. student exam-period support)	

4.2. Initial Go-to-Market Strategy

FoodLoop employs a community-anchored, phased launch focusing on high-density activity in pilot zones to trigger network effects before scaling nationally.

1. Phased Rollout Schedule

- Phase 1 — Pilot (Months 1–3): Concentrated launch in the Colombo Metropolitan Area. Focuses on "Anchor Institutions" (2–3 restaurants, 1–2 supermarkets, 1 national university, and 3 NGOs) to solve the "cold start" problem.
- Phase 2 — Expansion (Months 4–6): Extension into the Western Province (Gampaha, Kalutara). Focus on onboarding the wedding catering sector and one major national supermarket chain.
- Phase 3 — National Rollout (Months 7–12): Expansion to Kandy, Galle, Jaffna, and Negombo. Formal launch of the Premium Enterprise CSR Suite to coincide with corporate reporting cycles.

2. Early Adoption & Marketing Tactics

- Zero-Cost Entry: Free onboarding for donors and NGOs during the pilot to remove financial friction and build immediate supply/demand.

- Founding Partner Program: Exclusive "Founding Partner" status and permanent profile badges for early restaurant and retail adopters.
- Student Ambassador Program: Recruiting university students as advocates and volunteer drivers in exchange for enhanced subsidy access and platform rewards.
- Impact-Driven Content: Using the Live Impact Map and real human stories (video/social media) to provide "living proof" of social and environmental contributions.

4.3. Partnerships & Funding Strategies

FoodLoop is designed for financial resilience, using a hybrid model that combines internal revenue, non-dilutive grants, and strategic impact investment to support its transition from a pilot to a regional circular economy leader.

1. Internal Revenue Streams (Self-Sustainability)

- Marketplace Commissions: A volume-driven fee on every C2C/B2C sale by home-based entrepreneurs.
- Retail Integration Fees: Recurring B2B subscriptions for supermarkets utilizing the AI Dynamic Discounting and Demand Sensing engines.
- Premium CSR Subscriptions: Tiered annual packages for corporate clients requiring blockchain-verified ESG data and live widgets.
- Sponsored Impact Campaigns: Corporate-funded food drives (e.g., student exam-period support) that amplify social impact while generating revenue.

2. Strategic Funding Sources

- Grants & Accelerators: Targeting international bodies (UNDP, FAO, WFP Innovation Accelerator) whose goals align with SDG 2 (Zero Hunger) and SDG 12 (Responsible Consumption).
- Impact Investment: Engaging the Lanka Angel Network for proof-of-concept scaling, followed by regional Social Impact Venture Capital (e.g., Omidyar Network) for Phase 3 national rollout.
- Tech Infrastructure Credits: Leveraging AWS Activate (up to \$100k in credits) and Google for Startups to offset cloud, AI, and blockchain hosting costs during early development.

5. Implementation

5.1. Tech Stack (Overview)

Frontend	React.js
Backend	Node.js / Express.js, FastAPI
Database	Mongo DB
LLM Service	Groq
Authentication Mechanism	JWT (JSON Web Token) Authentication
Real Time Updates	Socket.io
Block Chain	Polygon, Solidity, Ethers.js
Map Library	Leaflet.js
Deployment	Frontend :- Vercel Backend :- Render

5.2. Impact on Stakeholders

1. Commercial Stakeholders (Supermarkets & Restaurants)

- **Cost Reduction:** Businesses save on expensive large-scale waste collection fees by redirecting surplus food to the platform.
- **Revenue Recovery:** AI-driven "Daily Moving Products" suggestions help retailers recover potential losses through real-time discount triggers.
- **Verified Sustainability:** The Polygon blockchain provides immutable proof of social impact, protecting brands from "greenwashing" accusations.

2. Social Stakeholders (NGOs & Vulnerable Groups)

- **Dignified Access:** A tiered "Social Aid" system provides low-income users and students priority access to free or subsidized high-quality food.
- **Logistical Efficiency:** AI Smart Routing ensures food arrives fresh while minimizing the time and fuel burden on charitable organizations.

3. Individual Marketplace Users (Donors & Customers)

- **Low-Friction Giving:** Households and small donors use a simple "Snap & List" interface to easily contribute to the circular economy.
- **Affordability:** Budget-conscious customers gain access to high-quality surplus food at significant discounts to help manage daily living costs.

4. The Logistics Ecosystem (Volunteer Drivers)

- **Precision Volunteering:** The Smart Routing Engine matches volunteers to deliveries based on their specific vehicle type to prevent wasted time.

- **Blockchain Incentives:** Volunteers earn verified "Impact Points" and digital certificates on the blockchain, which can be used for career development or gamification.

5.3. Onion Diagram

FoodLoop - Onion Diagram



5.4. Real-World Scenarios & User Journey

These scenarios demonstrate how **FoodLoop** integrates AI, blockchain, and real-time logistics to solve the specific pain points of diverse stakeholders:

- **The Restaurant Donor (Operations Manager):** When faced with unsold portions at closing, the manager uses **AI Snap & List** to list food in under 60 seconds. After safety clearance, an NGO claims the food, and the manager receives a **Blockchain Impact Receipt** confirming the social impact by morning.
- **The Supermarket Manager (Branch Manager):** Following a **Demand Sensing** alert about low foot traffic, the manager approves **AI Dynamic Discounting** for near-expiry items. Any unsold stock is automatically converted into a donation for an NGO, and a verified **ESG Report** is generated with a single click.
- **The Home-Based Seller (Entrepreneur):** Transitioning from unprotected social media sales, the entrepreneur gains consumer trust through a **Verified Seller Badge** and AI safety screening. Secure in-app payments protect her revenue, while weekend surplus can be instantly converted into donations.
- **The Vulnerable User (University Student):** Facing financial insecurity, the student accesses a private **"Subsidized Offers"** section with significant discounts and proximity alerts for donations. She can claim meal packs digitally and discreetly, ensuring dignified access without the stigma of public queues.
- **The Corporate CSR Manager (Head of Sustainability):** Replacing manual spreadsheets, the manager uses a **Premium Enterprise CSR Suite** populated with real-time blockchain data. This allows for the generation of formatted **ESG Impact Reports** and the use of live widgets to provide transparent proof of sustainability commitments.

6. Project Management

6.1. Project Management Approach

For the development of FoodLoop, our team will adopt the Agile methodology, specifically utilizing the Scrum framework. This approach is selected to manage the inherent complexities of integrating AI, blockchain, and real-time logistics while ensuring the platform remains responsive to stakeholder feedback.

1. Task Organization & Execution

The project will be broken down into two-week Sprints, allowing for the incremental delivery of functional features.

- **Product Backlog:** All high-level requirements—such as the AI "Snap & List" engine, Polygon blockchain integration, and the Dynamic Discounting module—will be documented and prioritized in a master backlog.
- **Sprint Planning:** At the start of each cycle, the team will select high-priority tasks to move into the Sprint Backlog for immediate development.
- **Daily Stand-ups:** Brief daily meetings will be held to discuss progress, synchronize activities, and identify any "blockers" that might impede development.

2. Collaboration & Progress Tracking

To ensure accountability and transparency across the multi-disciplinary team, we will utilize a suite of digital collaboration tools:

- **Project Management (Jira/Trello):** We will use digital Kanban boards to track the status of every task (e.g., "To Do," "In Progress," "Testing," "Done").
- **Version Control (GitHub):** All code for the React.js frontend, Node.js/FastAPI backend, and Solidity smart contracts will be managed via GitHub to facilitate seamless collaboration and peer reviews.
- **UI/UX Synchronization (Figma):** Designers and developers will remain aligned through our shared Figma workspace to ensure the "Privacy-by-Design" interface is implemented accurately.

3. Quality Assurance & Review

- **Sprint Review:** At the end of each sprint, we will demonstrate the newly developed features (e.g., a working "Impact Receipt" or the Live Impact Map) to ensure they meet our technical and ethical standards.
- **Sprint Retrospective:** The team will analyze the previous work cycle to identify process improvements, ensuring the development of the Minimum Viable Product (MVP) progresses efficiently toward the Phase 1 pilot launch.

6.2. Project Timeline & Milestones

The development of the **FoodLoop** Minimum Viable Product (MVP) is structured into a fast-tracked, five-week intensive cycle to ensure a high-quality launch by **May 10th**. This timeline prioritizes the core "Zero-Friction" AI and Blockchain transparency features essential for our Phase 1 pilot in Colombo.

Phase	Duration	Major Milestones	Expected Deliverables
1. Planning & Design	Week 1	Finalize System Architecture & UI/UX.	Finalized ERD, API Documentation, and high-fidelity Figma prototypes.
2. Backend Development	Weeks 2-3	Database setup & AI/Blockchain integration.	Functional Node.js / FastAPI server, Polygon smart contracts, and AI Snap & List API.
3. Frontend Development	Weeks 3-4	Client-side UI & State Management integration.	Responsive React.js web platform and mobile-ready interface with Leaflet.js maps.
4. Testing & QA	Week 5 (Early)	Security audits & End-to-End (E2E) testing.	Debugged MVP, "Freshness Guardian" computer vision validation, and JWT security report.

Phase	Duration	Major Milestones	Expected Deliverables
5. Final Deployment	By May 10th	Production launch on Vercel and Render.	Live FoodLoop MVP ready for anchor institution onboarding.

6.3. Team Roles & Responsibilities

To ensure the successful delivery of **FoodLoop** by the **May 10th** deadline, tasks are distributed based on specialized technical strengths, ensuring clear accountability and streamlined collaboration.

1. Individual Roles & Core Responsibilities

- **Ahamed Umar Jiffry – Team Leader & Full-Stack Developer**
 - **Project Oversight:** Managing the Agile Scrum process, Sprint Planning, and overall project timeline to meet the May 10th milestone.
 - **AI & Logic Integration:** Developing the "Snap & List" computer vision models and the "Demand Sensing" engine using FastAPI.
 - **Security Architecture:** Implementing JWT-based authentication and "Privacy-by-Design" data encryption for vulnerable users.
- **Jazeel Jaufer – UI/UX & Frontend Engineer**
 - **Interface Development:** Building the responsive React.js web platform and mobile interface as outlined in the Figma prototypes.
 - **Mapping & Real-Time UI:** Integrating Leaflet.js for the Live Impact Map and Socket.io for real-time donation alerts.
 - **Gamification Logic:** Implementing the digital badge system and donor milestone trackers.
- **Jegatheesan Risikesan – Backend & Blockchain Developer**
 - **Smart Contract Development:** Writing and deploying Solidity contracts on the Polygon network for immutable "Impact Receipts".
 - **Database Management:** Architecting the MongoDB schema to manage marketplace transactions and user eligibility data.
 - **Logistics Engine:** Developing the AI Smart Routing logic to match food volumes with optimal vehicle types

7. Additional and Supportive Materials

1) Wireframes & UI/UX Design :-

- Figma Link :- <https://www.figma.com/design/IAMcEJdEL9xekkJI2R3mw7/foodloop-copy?node-id=178-21829&m=dev>

8. Team



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